

Benchmark run full detail

Run ID: 20260428T201035Z_pure_full

Generation model: gemini / gemini-3.1-pro-preview

Standard validator: gemini / gemini-2.5-flash

Layer 1 uses sentence-transformer greedy one-to-one matching at threshold 0.55. Layer 2 uses a checklist-based Gemini judge with strict and weighted scoring.

Document: 0000_cctns

Part A - Document header

Document ID	0000_cctns
Generation model	gemini / gemini-3.1-pro-preview
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	100
Generated requirement count	67
Matched count	63
Clarification rounds	requested 2 / used 2
Clarification chunks	2
Semantic P / R / F1	0.9403 / 0.6300 / 0.7545
LLM strict P / R / F1	0.5224 / 0.3500 / 0.4192
LLM weighted P / R / F1	0.6940 / 0.4650 / 0.5569
Evidence / Propositions / Gap	103 / 67 / 4
Grounded after rewrite	67

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?
[USER] This project is based on the 0000 - cctns system, and the goal is to capture the main requirements and constraints for document 0000_cctns.
[SYSTEM] Got it. Since we are capturing the core requirements for the CCTNS system, could you describe the most frequent day-to-day tasks users will perform and how we might streamline those specific workflows to require as few steps as possible?
[USER] To answer your question about streamlining workflows, our main goal is efficiency. For the things our team does all the time, we really need those frequently-executed system transactions to be designed so they can be completed with a small number of interactions. We want to avoid having them click through endless screens just to get their daily work done.
[SYSTEM] That makes perfect sense for keeping things efficient. As users are moving quickly through these streamlined screens, how can we best design the layout—like the placement of page titles and navigation menus—so they always know exactly where they are in the system without getting lost?

[USER] To keep people from getting lost, we definitely need page titles placed in a consistent location on the different pages. Also, whenever navigation components or overviews are shown in conjunction with associated content, we have to maintain consistency between the navigation component and the content shown. It should clearly indicate where the user is right in the navigation component. When designing these visual cues, color should be used with care. We have to take into account human capabilities and restrictions in perceiving color, so it shouldn't be used as the only means of conveying information. Finally, to avoid link overload, within-page links are shown with dashed rather than solid underlines. If we have text pages containing large proportions of links, they need to be formatted so that the presence of links does not impede the readability of the text.

[SYSTEM] Those are excellent points about ensuring readability and accommodating color perception. Building on that accessibility focus, how should we handle non-text elements like images or videos so that users who rely on text-based alternatives can still get all the information they need?

[USER] To answer your question, it is crucial that all non-text media objects, such as graphical images or video, are provided with alternative equivalent textual descriptions and/or with equivalent text-based functionality. That way, users relying on screen readers do not miss anything. While we are on the topic of accessibility, we also need to ensure input device independence. The user interfaces should be designed to allow activation of controls by a variety of input devices, not just a standard mouse. For our overall approach to ICT accessibility, we will use ISO 9241-20 as the standard for guidance. And specifically for making sure the text itself is easy to read, ISO 9241-303 needs to be consulted for screen text legibility requirements.

[SYSTEM] Those legibility standards are great to have in place. Beyond just making the text readable, if a user gets stuck or isn't sure how to complete a specific action on a screen, what kind of contextual help or guidance should the system offer them?

[USER] When someone gets stuck, we really need a dedicated Help Module. It has to offer detailed context-sensitive help material for all the possible actions and scenarios they might encounter. And this needs to be available across all user interfaces in the application, so they are never left guessing. Also, going back to the legibility you mentioned, it is crucial that text should be able to be resized by the user. They should be able to do this easily using functions provided by the user agent or other appropriate means. Finally, if the system is tailoring the experience for them, we need to be transparent. It should be made evident to the user when individualization and/or adaptation are used, so they understand why the interface looks or behaves a certain way.

[SYSTEM] It makes a lot of sense to allow users to resize text for better legibility. To further help them digest information efficiently, how might we structure the content—perhaps using clear headings, bulleted lists, or highlighted keywords—so they can quickly skim the pages to find what they need?

[USER] Yes, exactly. To really support fast skimming of text, we definitely need to use clear links, bulleted lists, highlighted keywords, and logical headings, along with short phrases and sentences. Also, while they are reading, horizontal scrolling should be avoided wherever possible because it really disrupts the flow. But it's not just about how the page looks; getting to the information is just as important. We want to minimize the number of navigation steps needed to reach a certain piece of content. At the same time, we know people look for things in different ways, so we need to make sure different mental models, navigation strategies and tasks of the user are taken into account. To do that effectively, we should be offering alternative access paths for navigating to a specific unit of content, which will really help support those different navigation strategies.

[SYSTEM] Since we're offering those alternative access paths and trying to minimize navigation steps, how should we handle linking back to the home page or other key landmark pages so users can easily reorient themselves if they navigate deep into the content?

[USER] To answer your question directly, we need every single page to have a link leading to the home page of the application, or at least to a landmark page that is easy to recognize for the user. That way they never get lost. To help them navigate even better, we want them to be able to personalize their workspace, so customisation should include, but need not be limited to, changes like adjusting the menu contents. Also, regarding the overall layout, if frames are used, care should be taken to avoid possible problems—especially those involving the use of the back button, bookmarking of pages, or scrolling of information. Finally, we want to get users into the content quickly, so splash screens should be avoided unless they provide useful content or feedback about the application state to the user.

[SYSTEM] Building on your points about clean layouts and getting users straight to the content, how should we determine the ideal length for individual pages, and how might we use empty background space or text-only alternatives to ensure the information remains easy to skim?

[USER] For page lengths, there isn't a strict rule. Instead, the length of a page should be selected to support the primary purpose and use of the page. As for the empty areas, any space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page. If the visual layout is ever an issue for accessibility or preference, alternatively, the user should be provided with an equivalent alternative text-only page. On top of that, to help users find what they need, we definitely need a separate navigation overview, such as a site map, showing the structure of the site in an overview form.

[SYSTEM] That site map will definitely help users grasp the overall structure. As we build out that overview and the main menus, how should we choose the terminology for our navigation links to ensure the labels are instantly familiar to users based on their prior experience?

[USER] For the terminology, we really need to label our navigation links—particularly links representing the main navigation structure—with terms that are familiar to the user. We should draw on their general knowledge, their prior experience in the application domain, or their experience of using other systems so they instantly know where to click. While we're designing those menus, we also have to ensure that navigation links are clearly distinguishable from controls activating some action. Users shouldn't have to guess if a click will take them to a new page or trigger a process. Also, please make sure the quality of textual content with respect to spelling and grammar is sufficient so as not to impede readability; bad typos just ruin the experience. Finally, the entire user interface needs to be suitable for users with special needs, so accessibility has to be a priority.

[SYSTEM] Building on your points about clear navigation links and overall accessibility, how should we indicate to users exactly where they currently are within the site's overall structure as they move from page to page?

[USER] To answer your question directly, we need to make sure that each presentation segment, whether that is a page or a window, provides the user with a clear and sufficient indication of where they are in the navigation structure. They should always understand their current segment position with respect to the overall structure. While we are talking about moving around, it is also really important that any links that open new browser windows or pop-up windows are clearly marked so users are not surprised. Beyond just navigation, we want to make sure the actual information is easy to digest. The reading and understanding of the textual content on the screen should be supported by suitable means. Ideally, we would see things like the use of short sentences, the division of the text into shorter chunks, or the presentation of content items in the form of bullet points. Finally, if they do get stuck, the support solution should be accessible to the users both from within the application and also outside the application through a browser interface.

[SYSTEM] Building on your point about keeping users oriented and allowing them to personalize their workspace, if the system adapts content based on different user profiles, how should we clearly indicate to the user which profile is currently active?

[USER] That is a great question. Whenever the platform is individualizing or adapting content based on predefined user profiles or user-specified profiles, it is crucial that the profile currently used is made evident to the person looking at the screen. They should never have to guess why they are seeing certain things. Also, on a related note about how we display this content, if frames are used in the interface, we absolutely need an alternative way of presenting relevant information without frames. Some of our users might rely on accessibility tools that struggle with framed layouts, so having that fallback is really important to us.

[SYSTEM] Building on the idea of different user profiles interacting with the system, how should we track their access to cases and related documents behind the scenes to ensure those activity records remain completely unalterable and are kept for the entire life of the case?

[USER] To answer your question about tracking, we need to make sure that any access to cases, and all other activities involving the cases and related documents or data, are stored in the audit trail. This is crucial to ensure legal admissibility and to assist in data recovery if something goes wrong. When we say these records need to be 'unalterable,' we mean exactly that: the audit trail data cannot be modified in any way or deleted by any user. As for retention, we have to maintain the audit trail for as long as required, which means keeping it at least for the life of the case to which it refers. Finally, we will sometimes need to pull this data out for review, so we need the ability to export audit trails for specified cases, but doing so must happen without affecting the audit trail stored by the System.

[SYSTEM] Since we just discussed how strict the system needs to be about tracking user access and preventing anyone from altering the audit logs, how should we handle the process of assigning or changing user access rights, privileges, and passwords to ensure only authorized super-users can make those security updates?

[USER] That's a critical point. When it comes to managing who can do what, we need to be absolutely strict. Any changes to security attributes for groups or users—which includes things like access rights, security level, privileges, password allocation and management—need to be restricted so they are made only by super-user. We can't have regular users messing with those settings. Also, while we're talking about system management and tracking, we want to make sure there's an interface for the user to log any defects or enhancement requests on the application and track thereafter. It helps us keep on top of issues and improvements.

[SYSTEM] Building on your point about giving users a way to log application issues, let's talk about preventing and handling everyday data entry errors. How should we utilize persistent, user-definable default values to streamline input, and what specific details should our error messages include to help users quickly resolve any mistakes they do make?

[USER] To speed up our workflows, we definitely want to allow persistent defaults for data entry where desirable. It shouldn't just be standard system defaults either; we need those to include user-definable values so individuals can tailor the inputs to their daily work. Now, when mistakes do happen, ease of use is our main priority. All error messages produced by the system have to be meaningful so they can be appropriately acted upon by the users who are likely to see them. Whether these messages are shown on the pages themselves or on special error pages, the content should clearly state the reason why the error occurred. And whenever possible, it needs to spell out the actions the user can take to resolve the error, rather than just giving them a dead end.

[SYSTEM] Following up on how we handle data entry errors and guide users to fix them, when citizens are registering their complaints, how should the system validate the facts and evidence they submit to ensure the police have reliable information for their investigations, and are there any specific international standards we should consult to guide this validation process?

[USER] To answer your first point, the basic workflow is that citizens can register their complaints with the police. We don't expect the software to do all the heavy lifting on validation itself; rather, based on the evidence, facts and following investigation, the police will take the complaint forward. As for how we guide that overall process and ensure reliability, we definitely want to ensure that the International Standards mentioned below are consulted for guidance.

[SYSTEM] Since we're consulting international standards for how citizens and police handle complaints, how should we approach the design of the interfaces to ensure they support multiple languages and can be customized or integrated with any specialist software the police might need?

[USER] That is a great question. First off, we definitely need to support a multilingual interface so different communities can use it easily. We also want the screens to be flexible, so the interfaces should be made customizable or user-configurable to the extent possible. Since the police have their own tools, the setup has to be compatible with specialist software that may be used, and it needs to follow appropriate interface guidelines. Whether someone is filing a complaint or managing the backend, we need End User and Administrator functions which are easy to use and intuitive throughout. Lastly, reliability is huge for us—we need the system designed in a manner that operational data is not lost in case of any failure of equipment or communication network.

[SYSTEM] Building on your point about preventing data loss during network or equipment failures, how should the system's interfaces interact with the specialist police software to ensure information is safely queued and synced if a connection temporarily drops?

[USER] To answer your question about handling those drops, our broad guidelines for system functionality really come down to this: we need it designed in a manner that operational data is not lost in case of any failure of equipment or communication network. It just has to hold onto that information safely until the connection is back. Also, while we are discussing the interface itself, we need to make sure it supports a multilingual interface so that different users can work with it comfortably.

[SYSTEM] Building on your requirement to keep data safe during equipment or network failures, I would like to clarify our targets for the system's overall availability. Over a typical three-month period, how much total time can we afford for planned maintenance versus unplanned outages, and is there a limit to the number of separate unplanned incidents the police operations can tolerate?

[USER] Police operations run around the clock, so availability is critical for us. Looking at a <rolling three month period>, we need to schedule things carefully so that planned downtime doesn't exceed <xx> hours. For unexpected issues, the total unplanned downtime shouldn't exceed <xx hours/minutes> over that same <rolling three month period>. It's also really disruptive if the system keeps dropping, so the actual number of incidents of unplanned downtime can't exceed <x> per <rolling three month period>. Finally, if the worst happens and we experience any software or hardware failure, we absolutely need to be able to restore the System—complete with inline synchronization—within no more than <xx> hours.

[SYSTEM] Now that we've established the strict availability and recovery targets for keeping the operational data safe, let's talk about who gets to see that data. When someone searches for a case they don't have full permissions for, should the system hide it entirely from the results, or just show basic metadata like the title, and will users need the ability to grant case access to specific colleagues or groups?

[USER] To answer your question about searches, if someone runs a quick or advanced search, we absolutely never want to include in the search result list any record which the user does not have the right to access. However, we need some flexibility depending on how we set things up. If a user requests access to, or searches for, a case which he does not have the right to access, the exact response needs to be selectable at configuration time—for example, having the option to just display title and metadata instead of the full file. Regarding sharing, yes, we definitely want to allow a user to stipulate which other users or groups can access cases. More broadly, we need solid role-based control for the functionality within the system to manage all these permissions. And since our staff often wear multiple hats, it's crucial that we allow a user to be a member of more than one group.

Part C - Extracted requirements by category

Functional

- FR-001** The system must allow users to personalize their workspace, so customisation should include, but need not be limited to, changes like adjusting the menu contents.
- FR-002** The system must allow the export of audit trails for specified cases without affecting the audit trail stored by the System.

- FR-003** The system must provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.
- FR-004** The system must allow persistent defaults for data entry where desirable, including user-definable values.
- FR-005** The system must allow citizens to register their complaints with the police.

Non-Functional

- NFR-001** Frequently-executed system transactions must be designed so they can be completed with a small number of interactions.
- NFR-002** The system must place page titles in a consistent location on different pages.
- NFR-003** The system must maintain consistency between navigation components or overviews and the associated content shown.
- NFR-004** The system must clearly indicate where the user is right in the navigation component.
- NFR-005** The system must not use color as the only means of conveying information.
- NFR-006** The system must show within-page links with dashed rather than solid underlines to avoid link overload.
- NFR-007** The system must format text pages containing large proportions of links so that the presence of links does not impede the readability of the text.
- NFR-008** The system must provide all non-text media objects, such as graphical images or video, with alternative equivalent textual descriptions and/or with equivalent text-based functionality.
- NFR-009** The system must allow activation of controls by a variety of input devices, not just a standard mouse.
- NFR-010** The system must use ISO 9241-20 as the standard for guidance for the overall approach to ICT accessibility.
- NFR-011** The system must consult ISO 9241-303 for screen text legibility requirements.
- NFR-012** The system must provide a dedicated Help Module offering detailed context-sensitive help material for all the possible actions and scenarios they might encounter.
- NFR-013** The system must make the Help Module available across all user interfaces in the application.
- NFR-014** The system must allow text to be resized easily by the user using functions provided by the user agent or other appropriate means.
- NFR-015** The system must make it evident to the user when individualization and/or adaptation are used.
- NFR-016** The system must use clear links, bulleted lists, highlighted keywords, and logical headings, along with short phrases and sentences to support fast skimming of text.
- NFR-017** The system must avoid horizontal scrolling wherever possible.
- NFR-018** The system must minimize the number of navigation steps needed to reach a certain piece of content.
- NFR-019** The system must offer alternative access paths for navigating to a specific unit of content.
- NFR-020** The system must include on every single page a link leading to the home page of the application, or at least to a landmark page that is easy to recognize for the user.
- NFR-021** The system must avoid possible problems, especially those involving the use of the back button, bookmarking of pages, or scrolling of information, if frames are used.
- NFR-022** The system must avoid splash screens unless they provide useful content or feedback about the application state to the user.
- NFR-023** The system must select the length of a page to support the primary purpose and use of the page.
- NFR-024** The system must use any space filled only with the background color in such a way that it does not impair the visual skimming of the page.

NFR-025	The system must provide the user with an equivalent alternative text-only page if the visual layout is ever an issue for accessibility or preference.
NFR-026	The system must provide a separate navigation overview, such as a site map, showing the structure of the site in an overview form.
NFR-027	The system must label navigation links with terms familiar to the user based on general knowledge, prior experience in the application domain, or experience using other systems.
NFR-028	The system must ensure that navigation links are clearly distinguishable from controls activating some action.
NFR-029	The system must ensure the quality of textual content with respect to spelling and grammar is sufficient so as not to impede readability.
NFR-030	The system must ensure the entire user interface is suitable for users with special needs.
NFR-031	The system must ensure that each presentation segment, whether that is a page or a window, provides the user with a clear and sufficient indication of where they are in the navigation structure.
NFR-032	The system must clearly mark any links that open new browser windows or pop-up windows so users are not surprised.
NFR-033	The system must support the reading and understanding of the textual content on the screen by suitable means, such as the use of short sentences, the division of the text into shorter chunks, or the presentation of content items in the form of bullet points.
NFR-034	The system must make the support solution accessible to users both from within the application and outside the application through a browser interface.
NFR-035	The system must make evident to the person looking at the screen the profile currently used whenever individualizing or adapting content based on predefined user profiles or user-specified profiles.
NFR-036	The system must provide an alternative way of presenting relevant information without frames if frames are used in the interface.
NFR-037	The system must store any access to cases, and all other activities involving the cases and related documents or data, in the audit trail.
NFR-038	The system must ensure the audit trail data cannot be modified in any way or deleted by any user.
NFR-039	The system must restrict changes to security attributes for groups or users—including access rights, security level, privileges, password allocation and management—so they are made only by super-user.
NFR-040	The system must produce meaningful error messages that clearly state the reason why the error occurred.
NFR-041	The system must spell out the actions the user can take to resolve the error in error messages whenever possible.
NFR-042	The system must consult the mentioned International Standards for guidance on the complaint validation process.
NFR-043	The system must support a multilingual interface.
NFR-044	The system must make interfaces customizable or user-configurable to the extent possible.
NFR-045	The system must follow appropriate interface guidelines for compatibility with specialist software.
NFR-046	The system must provide End User and Administrator functions which are easy to use and intuitive throughout.
NFR-047	The system must be designed in a manner that operational data is not lost in case of any failure of equipment or communication network.
NFR-048	The system must ensure operational data is not lost in case of any failure of equipment or communication network, holding onto that information safely until the connection is back.

NFR-049	The system must ensure total unplanned downtime does not exceed <xx hours/minutes> over a <rolling three month period>.
NFR-050	The system must ensure the number of incidents of unplanned downtime does not exceed <x> per <rolling three month period>.
NFR-051	The system must be able to restore complete with inline synchronization within no more than <xx> hours after any software or hardware failure.
NFR-052	The system must never include in search result lists any record which the user does not have the right to access.
NFR-053	The system must allow the exact response to be selectable at configuration time if a user requests access to, or searches for, a case which he does not have the right to access, for example, having the option to just display title and metadata instead of the full file.
NFR-054	The system must allow a user to stipulate which other users or groups can access cases.
NFR-055	The system must provide role-based control for the functionality within the system to manage permissions.
NFR-056	The system must allow a user to be a member of more than one group.
NFR-057	The entire user interface must be suitable for users with special needs.
NFR-058	The quality of textual content with respect to spelling and grammar must be sufficient so as not to impede readability.
NFR-059	Color must be used with care when designing visual cues.
NFR-060	Any links that open new browser windows or pop-up windows must be clearly marked.

Data

DATA-001	The system must maintain the audit trail for as long as required, keeping it at least for the life of the case to which it refers.
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Interfaces

IF-001	The system must be compatible with specialist software that may be used by the police.
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Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
Registration Citizens can register their complaints with police and then based on the evidence, facts and following investigation, police shall take the complaint forward.	✓	The system must allow citizens to register their complaints with the police.
Help Module The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	✓	The system must provide a dedicated Help Module offering detailed context-sensitive help material for all the possible actions and scenarios they might encounter.
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	✓	The system must make the Help Module available across all user interfaces in the application.
The solution should provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.	✓	The system must provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.
The solution should send alerts (e.g., email, SMS) to the user if the user chooses to whenever any action has been taken on the alert.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The solution should enable the user to track the submitted defect or enhancement request.	✗	— not recovered
The solution should enable the help-desk user to view the reports on the submitted defects or enhancement requests category-wise, status-wise, and age- wise.	✗	— not recovered
The support solution should be accessible to the users both from within the application and also outside the application through a browser interface.	✓	The system must make the support solution accessible to users both from within the application and outside the application through a browser interface.
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	✓	The system must ensure the audit trail data cannot be modified in any way or deleted by any user.
Administrative parameters The word “unalterable” is to mean that the audit trail data cannot be modified in any way or deleted by any user;	✗	— not recovered
it may be subject to re-department and copying to removable media if required, so long as its contents remain unchanged.	✗	— not recovered
Once the audit trail functionality has been activated, the System must track events without manual intervention, and store in the audit trail information about them.	✗	— not recovered
The System must maintain the audit trail for as long as required, which will be at least for the life of the case to which it refers.	✓	The system must maintain the audit trail for as long as required, keeping it at least for the life of the case to which it refers.
The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	✓	The system must allow the export of audit trails for specified cases without affecting the audit trail stored by the System.
The System must be able to capture and store violations (i.e.	✗	— not recovered
The System must at a minimum be able to provide reports for actions on cases organised: By case;	✗	— not recovered
The System should be able to provide reports for actions on cases organised by workstation and (where technically appropriate) by network address.	✗	— not recovered
The System must allow the user to limit access to cases to specified users or user groups.	✗	— not recovered
The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	✓	The system must provide role-based control for the functionality within the system to manage permissions.
The System must allow only admin-users to set up user profiles and allocate users to groups.	✓	The system must allow a user to be a member of more than one group.
The System should allow a user to stipulate which other users or groups can access cases.	✓	The system must allow a user to stipulate which other users or groups can access cases.
The System must allow changes to security attributes for groups or users (such as access rights, security level, privileges, password allocation and management) to be made only by super-user.	✓	The system must restrict changes to security attributes for groups or users—including access rights, security level, privileges, password allocation and management—so they are made only by super-user.

Source requirement text	Matched?	Closest generated requirement text
If a user requests access to, or searches for, a case which he does not have the right to access, the System must provide one of the following responses (selectable at configuration time): display title and metadata;	✓	The system must allow the exact response to be selectable at configuration time if a user requests access to, or searches for, a case which he does not have the right to access, for example, having the option to just display title and metadata instead of the full file.
the most stringent) implies that the System must not include such cases in any count of search results;	✗	— not recovered
If a user performs a quick or advanced search, the System must never include in the search result list any record which the user does not have the right to access.	✓	The system must never include in search result lists any record which the user does not have the right to access.
If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.	✗	— not recovered
Any access to cases, and all other activities involving the cases and related documents or data should also need to be stored in the audit trail to ensure legal admissibility and to assist in data recovery.	✓	The system must store any access to cases, and all other activities involving the cases and related documents or data, in the audit trail.
Ease of Use All error messages produced by the System must be meaningful, so that they can be appropriately acted upon by the users who are likely to see them.	✓	The system must produce meaningful error messages that clearly state the reason why the error occurred.
The System must employ a single set of user interface rules, or a small number of sets to provide a familiar and common look and feel for the application.	✗	— not recovered
The System must be able to display several entities (cases, suspects) simultaneously.	✗	— not recovered
The interfaces must be made customizable or user-configurable to the extent possible.	✓	The system must make interfaces customizable or user-configurable to the extent possible.
Such configurations must be saved in the user profile.	✗	— not recovered
The System user interface must be suitable for users with special needs;	✓	The system must ensure the entire user interface is suitable for users with special needs.
that is, compatible with specialist software that may be used and with appropriate interface guidelines The System must provide End User and Administrator functions which are easy to use and intuitive throughout.	✓	The system must provide End User and Administrator functions which are easy to use and intuitive throughout.
The System must allow persistent defaults for data entry where desirable.	✓	The system must allow persistent defaults for data entry where desirable, including user-definable values.
These defaults should include: user-definable values;	✗	— not recovered
Frequently-executed System transactions must be designed so that they can be completed with a small number of interactions (e.g.	✓	Frequently-executed system transactions must be designed so they can be completed with a small number of interactions.
Where the System employs a graphical user interface, it must allow users to customise it.	✗	— not recovered
Customisation should include, but need not be limited to the following changes: menu contents;	✓	The system must allow users to personalize their workspace, so customisation should include, but need not be limited to, changes like adjusting the menu contents.
Usability The user interfaces should be designed to make them user-intuitive.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The user interfaces of the system should comply with Standard ISO 9241.	✓	The system must follow appropriate interface guidelines for compatibility with specialist software.
ICT accessibility: ISO 9241-20 shall be the standard for guidance on ICT accessibility.	✓	The system must use ISO 9241-20 as the standard for guidance for the overall approach to ICT accessibility.
Software accessibility ISO 9241-171 shall be the standard for guidance on software accessibility.	✗	— not recovered
User interfaces should meet its requirements and recommendations.	✓	The entire user interface must be suitable for users with special needs.
Content accessibility WCAG 1.0 shall be the standard used for guidance on content accessibility.	✗	— not recovered
Providing text equivalents for non-text media objects: All non-text media objects, such as graphical images or video, should be provided with alternative equivalent textual descriptions and/or with equivalent text-based functionality.	✓	The system must provide all non-text media objects, such as graphical images or video, with alternative equivalent textual descriptions and/or with equivalent text-based functionality.
Making navigation self-descriptive: Navigation should be designed to help users understand where they are, where they have been and where they can go next.	✗	— not recovered
Showing users where they are: Each presentation segment (page or window) should provide the user with a clear and sufficient indication of where he or she is in the navigation structure and of the current segment position with respect to the overall structure.	✓	The system must ensure that each presentation segment, whether that is a page or a window, provides the user with a clear and sufficient indication of where they are in the navigation structure.
Offering alternative access paths: Alternative access paths for navigating to a specific unit of content should be offered to support different navigation strategies.	✓	The system must offer alternative access paths for navigating to a specific unit of content.
Minimizing navigation effort: The number of navigation steps needed to reach a certain piece of content should be minimized as long as different mental models, navigation strategies and tasks of the user are taken into account.	✓	The system must minimize the number of navigation steps needed to reach a certain piece of content.
Splash screens should be avoided unless they provide useful content or feedback about the application state to the user.	✓	The system must avoid splash screens unless they provide useful content or feedback about the application state to the user.
If a splash screen is used, a navigation option to skip it should be offered.	✗	— not recovered
Avoiding opening unnecessary windows: Additional windows such as new browser windows or pop-up windows should only be opened if this supports the user's task.	✓	The system must clearly mark any links that open new browser windows or pop-up windows so users are not surprised.
Horizontal scrolling should be avoided wherever possible.	✓	The system must avoid horizontal scrolling wherever possible.
Designing for input device independence: User interfaces should be designed to allow activation of controls by a variety of input devices.	✓	The system must allow activation of controls by a variety of input devices, not just a standard mouse.
Making user interfaces robust: User interfaces should be designed to be as robust as possible in the face of changing technology.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Acceptable opening / download times: Application pages should be designed and implemented so that there are acceptable opening times and download times for the expected range of technical contexts of use (e.g.	✗	— not recovered
Minimizing user errors: Potential user errors as well as the effort needed to recover from errors should be minimized.	✗	— not recovered
Providing clear error messages: The content of error messages shown on the pages or special error pages should clearly state the reason why the error occurred and, if possible, actions the user can take to resolve the error.	✓	The system must spell out the actions the user can take to resolve the error in error messages whenever possible.
Making text resizable by the user: Text should be able to be resized by the user, using functions provided by the user agent or other appropriate means i.e.	✓	The system must allow text to be resized easily by the user using functions provided by the user agent or other appropriate means.
Text quality: The quality of textual content with respect to spelling and grammar should be sufficient so as not to impede readability.	✓	The quality of textual content with respect to spelling and grammar must be sufficient so as not to impede readability.
Writing style: The reading and understanding of the textual content on the screen should be supported by suitable means, including the use of short sentences, the division of the text into shorter chunks or the presentation of content items in the form of bullet points.	✓	The system must support the reading and understanding of the textual content on the screen by suitable means, such as the use of short sentences, the division of the text into shorter chunks, or the presentation of content items in the form of bullet points.
Supporting text skimming: Fast skimming of text should be supported by the provision of clear links, bulleted lists, highlighted keywords, logical headings, and short phrases and sentences.	✓	The system must use clear links, bulleted lists, highlighted keywords, and logical headings, along with short phrases and sentences to support fast skimming of text.
Readability of text: Text presented on the pages should be readable taking into account the expected display characteristics and spatial arrangement.	✗	— not recovered
ISO 9241-303 shall be consulted for screen text legibility requirements.	✓	The system must consult ISO 9241-303 for screen text legibility requirements.
Distinguishable within-page links: Within-page links should be clearly distinguishable from other links that lead to a different page.	✗	— not recovered
Within-page links are shown with dashed rather than solid underlines Avoiding link overload: Text pages containing large proportions of links should be formatted so that the presence of links does not impede the readability of the text.	✓	The system must show within-page links with dashed rather than solid underlines to avoid link overload.
Using familiar terminology for navigation links: Navigation links — particularly links representing the main navigation structure — should be labelled with terms that are familiar to the user, based on his/her general knowledge, prior experience in the application domain or experience of using other systems.	✓	The system must label navigation links with terms familiar to the user based on general knowledge, prior experience in the application domain, or experience using other systems.
Marking links opening new windows: Links that open new browser windows or pop-up windows should be clearly marked.	✓	Any links that open new browser windows or pop-up windows must be clearly marked.
Distinguishing navigation links from controls: Navigation links should be clearly distinguishable from controls activating some action.	✓	The system must ensure that navigation links are clearly distinguishable from controls activating some action.

Source requirement text	Matched?	Closest generated requirement text
space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page.	✓	The system must use any space filled only with the background color in such a way that it does not impair the visual skimming of the page.
Selecting appropriate page lengths The length of a page should be selected so as to support the primary purpose and use of the page.	✓	The system must select the length of a page to support the primary purpose and use of the page.
Using colour: Colour should be used with care, taking into account human capabilities and restrictions in perceiving colour, and not as the only means of conveying information.	✓	Color must be used with care when designing visual cues.
Color should never be the only means of coding.	✓	The system must not use color as the only means of conveying information.
Using frames with care: If frames are used, care should be taken to avoid possible problems, for example, those involving the use of the back button, bookmarking of pages, or scrolling of information.	✓	The system must avoid possible problems, especially those involving the use of the back button, bookmarking of pages, or scrolling of information, if frames are used.
Providing alternatives to frame-based presentation: If frames are used, an alternative way of presenting relevant information without frames should be provided.	✓	The system must provide an alternative way of presenting relevant information without frames if frames are used in the interface.
Providing alternative text-only pages: When style sheets and/or frames are turned off it should be possible for the user to read and understand the page;	✗	— not recovered
alternatively, the user should be provided with an equivalent alternative text-only page.	✓	The system must provide the user with an equivalent alternative text-only page if the visual layout is ever an issue for accessibility or preference.
Consistent page layout: Pages should be designed using consistent layout schemes, supporting the user in finding similar information at the same position on different pages.	✗	— not recovered
Placing title information consistently: Page titles should be placed in a consistent location on the different pages.	✓	The system must place page titles in a consistent location on different pages.
Observing principles of human perception When designing application pages, the general principles of human perception should be taken into account.	✗	— not recovered
The International Standards mentioned below shall be consulted for guidance.	✓	The system must consult the mentioned International Standards for guidance on the complaint validation process.
In addition, when designing multimedia information presentations, the design principles and recommendations described in ISO 14915-1 to ISO 14915-3 should be taken into account.	✗	— not recovered
Linking back to the home page or landmark pages: Each page should contain a link leading to the home page of the application or to a landmark page that is easy to recognize for the user.	✓	The system must include on every single page a link leading to the home page of the application, or at least to a landmark page that is easy to recognize for the user.
Providing a site map: A separate navigation overview such as a site map should be provided for application showing the structure of the site in an overview form.	✓	The system must provide a separate navigation overview, such as a site map, showing the structure of the site in an overview form.

Source requirement text	Matched?	Closest generated requirement text
Consistency between navigation components and content: If navigation components (or overviews) are shown in conjunction with associated content, consistency between the navigation component and the content shown should be maintained by indicating in the navigation component (e.g.	✓	The system must maintain consistency between navigation components or overviews and the associated content shown.
Placing navigation components consistently: Navigation components should be placed consistently on the pages or in the framesets in the pages of the application.	✓	The system must clearly indicate where the user is right in the navigation component.
Taking account of the users' tasks and information needs: When providing different access paths or navigation structures for different user groups, the tasks and information needs of these user groups should be taken into consideration.	✗	— not recovered
Making individualization and adaptation evident: It should be made evident to the user when individualization and/or adaptation are used.	✓	The system must make it evident to the user when individualization and/or adaptation are used.
Making user profiles evident: If predefined user profiles or user-specified profiles are used for individualizing or adapting content, the profile currently used should be made evident.	✓	The system must make evident to the person looking at the screen the profile currently used whenever individualizing or adapting content based on predefined user profiles or user-specified profiles.
Allowing users to see and change profiles: If user-specified profiles are used, users should be able to see, modify and delete that profile on demand.	✗	— not recovered
The user interfaces of the system should follow the guidelines specified under www.usability.gov System Availability The System must be available to users: from <xx:00> to <xx:00>;	✗	— not recovered
The planned downtime for the System must not exceed <xx> hours per <rolling three month period>.	✗	— not recovered
Unplanned downtime for the System must not exceed <xx hours/minutes> per <rolling three month period>.	✓	The system must ensure total unplanned downtime does not exceed <xx hours/minutes> over a <rolling three month period>.
The number of incidents of unplanned downtime for the System must not exceed <x> per <rolling three month period>.	✓	The system must ensure the number of incidents of unplanned downtime does not exceed <x> per <rolling three month period>.
In the event of any software or hardware failure, it must be possible to restore the System (with inline synchronization) within no more than <xx> hours.	✓	The system must be able to restore complete with inline synchronization within no more than <xx> hours after any software or hardware failure.
The System must be able to retrieve and display within 5-8 seconds the case which has been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✗	— not recovered
The System must be able to retrieve and display within 20 seconds the case which has not been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✗	— not recovered
The System be scaleable and must not have any features which would preclude use in small or large police stations, with varying numbers of cases handled.	✓	The system must be compatible with specialist software that may be used by the police.

Source requirement text	Matched?	Closest generated requirement text
The systems should be designed with the following broad guidelines: System Functionality The system should support multilingual interface The system should be designed in manner that operational data is not lost in case of any failure of equipment or communication network.	✓	The system must support a multilingual interface.

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.5224 / 0.3500 / 0.4192. Weighted P/R/F1: 0.6940 / 0.4650 / 0.5569.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	The System must maintain the audit trail for as long as required, which will be at least for the life of the case to which it refers.	The system must maintain the audit trail for as long as required, keeping it at least for the life of the case to which it refers.	The generated requirement faithfully covers the source requirement regarding audit trail retention period.
full	Frequently-executed System transactions must be designed so that they can be completed with a small number of interactions (e.g.	Frequently-executed system transactions must be designed so they can be completed with a small number of interactions.	The generated requirement faithfully covers the source requirement, preserving all critical details regarding frequently-executed transactions and interactions.
full	The System must allow changes to security attributes for groups or users (such as access rights, security level, privileges, password allocation and management) to be made only by super-user.	The system must restrict changes to security attributes for groups or users—including access rights, security level, privileges, password allocation and management—so they are made only by super-user.	The generated requirement faithfully rephrases the source, preserving all critical details and the core need.
full	Unplanned downtime for the System must not exceed <xx hours/minutes> per <rolling three month period>.	The system must ensure total unplanned downtime does not exceed <xx hours/minutes> over a <rolling three month period>.	The generated requirement faithfully covers the source requirement regarding total unplanned downtime and its duration over a specified period.
full	The number of incidents of unplanned downtime for the System must not exceed <x> per <rolling three month period>.	The system must ensure the number of incidents of unplanned downtime does not exceed <x> per <rolling three month period>.	The generated requirement faithfully covers the source requirement regarding the number of incidents of unplanned downtime and its frequency over a specified period.
full	The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	The system must allow the export of audit trails for specified cases without affecting the audit trail stored by the System.	The generated requirement faithfully covers the source requirement regarding the export of audit trails.
full	The interfaces must be made customizable or user-configurable to the extent possible.	The system must make interfaces customizable or user-configurable to the extent possible.	The generated requirement faithfully covers the source requirement, preserving all critical details regarding customizable interfaces.
full	In the event of any software or hardware failure, it must be possible to restore the System (with inline synchronization) within no more than <xx> hours.	The system must be able to restore complete with inline synchronization within no more than <xx> hours after any software or hardware failure.	The generated requirement faithfully covers the source requirement regarding system restoration time, including inline synchronization, after software or hardware failure.

Verdict	Source requirement	Generated requirement	Judge reason
full	ISO 9241-303 shall be consulted for screen text legibility requirements.	The system must consult ISO 9241-303 for screen text legibility requirements.	The generated requirement is identical to the source, only changing the modality from 'shall be consulted' to 'must consult' while preserving all critical details.
full	Providing a site map: A separate navigation overview such as a site map should be provided for application showing the structure of the site in an overview form.	The system must provide a separate navigation overview, such as a site map, showing the structure of the site in an overview form.	The generated requirement faithfully covers all critical details of the source requirement regarding providing a site map or navigation overview.
full	ICT accessibility: ISO 9241-20 shall be the standard for guidance on ICT accessibility.	The system must use ISO 9241-20 as the standard for guidance for the overall approach to ICT accessibility.	The generated requirement faithfully covers the source, preserving all critical details and adding only a minor, non-material clarification ('overall approach to').
full	Marking links opening new windows: Links that open new browser windows or pop-up windows should be clearly marked.	Any links that open new browser windows or pop-up windows must be clearly marked.	The generated requirement faithfully captures the source, changing 'should be' to 'must be' which is a common and acceptable strengthening in requirement recovery.
full	Text quality: The quality of textual content with respect to spelling and grammar should be sufficient so as not to impede readability.	The quality of textual content with respect to spelling and grammar must be sufficient so as not to impede readability.	The generated requirement is almost identical to the source, only strengthening the modality from 'should be' to 'must be' while preserving all critical details.
full	space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page.	The system must use any space filled only with the background color in such a way that it does not impair the visual skimming of the page.	The generated requirement faithfully captures the source, strengthening the modality and slightly generalizing 'space' to 'any space'.
full	The support solution should be accessible to the users both from within the application and also outside the application through a browser interface.	The system must make the support solution accessible to users both from within the application and outside the application through a browser interface.	The generated requirement faithfully covers the source, preserving all critical details.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Help Module The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	The system must make the Help Module available across all user interfaces in the application.	The generated requirement only covers the availability of the Help Module across all user interfaces, omitting the critical detail about the content being 'detailed context-sensitive help material for all possible actions and scenarios'.
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	The system must provide a dedicated Help Module offering detailed context-sensitive help material for all the possible actions and scenarios they might encounter.	The generated requirement omits the critical detail that the help material should be available 'on all user interfaces in the application'.
The solution should enable the user to track the submitted defect or enhancement request.	The system must provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.	The generated requirement adds the explicit need to 'log any defects or enhancement requests', whereas the source only specifies enabling the user to 'track the submitted' ones.

Source requirement	Generated requirement	Judge reason
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	The system must store any access to cases, and all other activities involving the cases and related documents or data, in the audit trail.	The generated requirement narrows the scope of audited entities to 'cases and related documents or data' and actions to 'access' and 'activities involving cases', losing the broader 'critical entities' and explicit 'create/read/update/delete' actions, and the 'unalterable' and 'automatically capturing' details.
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	The system must ensure the audit trail data cannot be modified in any way or deleted by any user.	The generated requirement only covers the immutability of the audit trail data, omitting the critical details about what information the audit trail must capture (actions on critical entities, user initiating actions).
Once the audit trail functionality has been activated, the System must track events without manual intervention, and store in the audit trail information about them.	The system must ensure the audit trail data cannot be modified in any way or deleted by any user.	The generated requirement focuses on the immutability and non-deletion of audit trail data, which is not explicitly stated in the source. The source focuses on automatic tracking and storage of events.
The System must be able to capture and store violations (i.e.	The system must allow citizens to register their complaints with the police.	The generated requirement narrows the scope of 'violations' to 'complaints with the police' and introduces specific roles (citizens, police) not present in the source.
The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	The system must allow a user to be a member of more than one group.	The generated requirement only covers the second part of the source requirement (users in multiple groups), omitting the role-based control aspect.
The System must allow only admin-users to set up user profiles and allocate users to groups.	The system must restrict changes to security attributes for groups or users—including access rights, security level, privileges, password allocation and management—so they are made only by super-user.	The generated requirement generalizes the actions from 'set up user profiles and allocate users to groups' to 'changes to security attributes' and uses 'super-user' instead of 'admin-users'.
The System should allow a user to stipulate which other users or groups can access cases.	The system must allow a user to stipulate which other users or groups can access cases.	The generated requirement changes the modality strength from 'should' to 'must'.